



Assessment and Demonstration

Knowledge and Technology Transfer

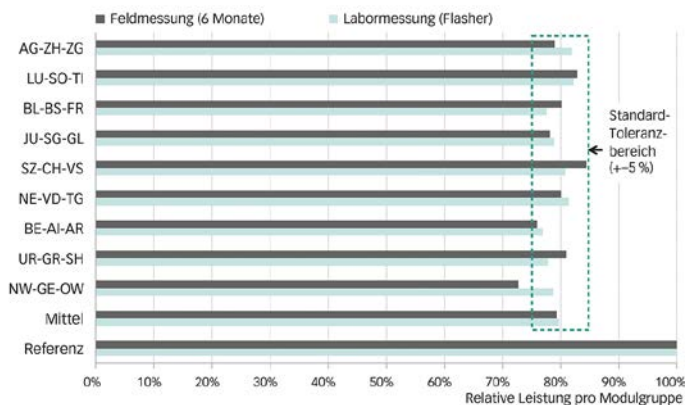
Besides dissemination of knowledge on visual assessment into the international scientific community, we have developed a technological concept for manufacturing coloured PV modules, which we transferred into a commercial product through support from technology transfer company ÜserHuus and Switzerland's leading glass manufacturing industrial company, Glas Trösch. Supplementary outreach activities inform and teach building professionals about this product and its applications through public talks, non-scientific publications and professional education courses.

Key words: SWISSPANEL SOLAR, patent, technology transfer

Target audience: building professionals, general public, glass and PV module manufacturing

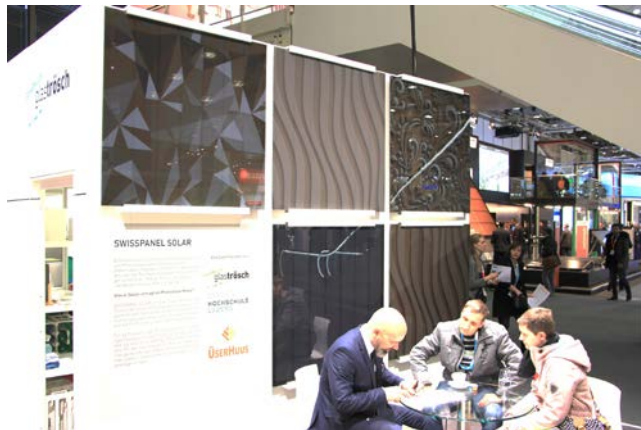
Patent and trademark

Several researchers aim at improving the visual acceptance of PV modules through colours, while limiting the efficiency losses through the shading effect of the colour. CSEM/EPFL's focus is on uniformly colouring the embedded encapsulant and modifying crystalline PV cells to optimally harness the resulting transmitted light spectrum. HSLU focuses on colouring the front glass using the digitally ceramic printing technology, enabling a diversity of designs through patterns with multiple colours for maximum visual integration into the built environment. We modified this technology to limit the dangerous hotspot effect that would naturally occur when PV cells would be covered with glass of different colours. In addition we have experimentally derived a relationship between print density and likely relative efficiency for selected print colours. Relative efficiency is the efficiency of a PV module with printed glass versus one with clear glass. As a result, we can simulate and specify the required colour for a desired relative efficiency or vice-versa. The basic principle was filed as patent [1] and trademark [2] in 2016. The image below shows the performance of such coloured PV modules. The desired relative efficiency was set to 80%, which resulted in specific print density settings for the required colours. The thus printed glass was then laminated into polycrystalline PV modules. Their resulting efficiency was measured in the lab and in the field and compared to identical PV modules with clear glass to determine the relative efficiency. The dotted box shows slight deviations from the 80% target for different designs (bars) and between lab and field measurement. However, deviations in the order of $\pm 5\%$ are not critical as they are within the standard deviation of typical production PV modules anyway.



Technology transfer and product development with industry

The patent marks a technology readiness level (TRL) of 5, as it is based on experimental field tests with prototypes. Through the support of the Technology Transfer company ÜserHuus, the TRL was moved to the final level 9, a full commercial application. The manufacturing was realized in collaboration with Glas Trösch, the leading glass processing company in Switzerland, which in turn led to the development of a new product SWISSPANEL SOLAR, which is a digitally printed glass optimized for PV application that can be supplied to any PV module manufacturer. This product was publicly launched at SWISSBAU 2018, the largest building fair in Switzerland [3], accompanied by a press release [4]. Testing for international certification is under way, financially supported by ÜserHuus and following our research partner SUPSI's research findings on compliance with the latest technological standards.



Swissbau 2018: Launch of SWISSPANEL SOLAR at Glas Trösch booth.

Dissemination to the general public

The product is aimed at architects and clients with projects that require the best visual integration of PV into the building envelope. In order to inform them of the opportunities afforded by coloured PV modules, we have placed a larger article in the leading Swiss magazine for building culture, providing a maximum outreach to architects, engineers, and planners as well as private and public clients. This paper informs of challenges with respect to colour, transparency, reflection and façade layout and the lessons learned from pilot and demonstration projects using the above product [5].

Professional education

Architects, PV manufacturers or planners who want to learn how to create coloured PV modules can attend a new professional education course at HSLU [6]. They learn about the meta-c-print method, experiences shared from Swiss research colleagues, product developers, makers of pilot- and demonstration projects and can manufacture their own PV module at CSEM/EPFL labs. This course is financially supported by EnergieSchweiz, and integrated into the Swissolar/SIA course system.

References

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- [2] Meta-c-print. IGE/IPT trademark 695585. <https://www.swissreg.ch/srclient/faces/jsp/trademark/sr30.jsp>
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- [6] Farbige PV Module. <https://www.hslu.ch/de-ch/technik-architektur/weiterbildung/fachkurse/farbige-photovoltaik-module>